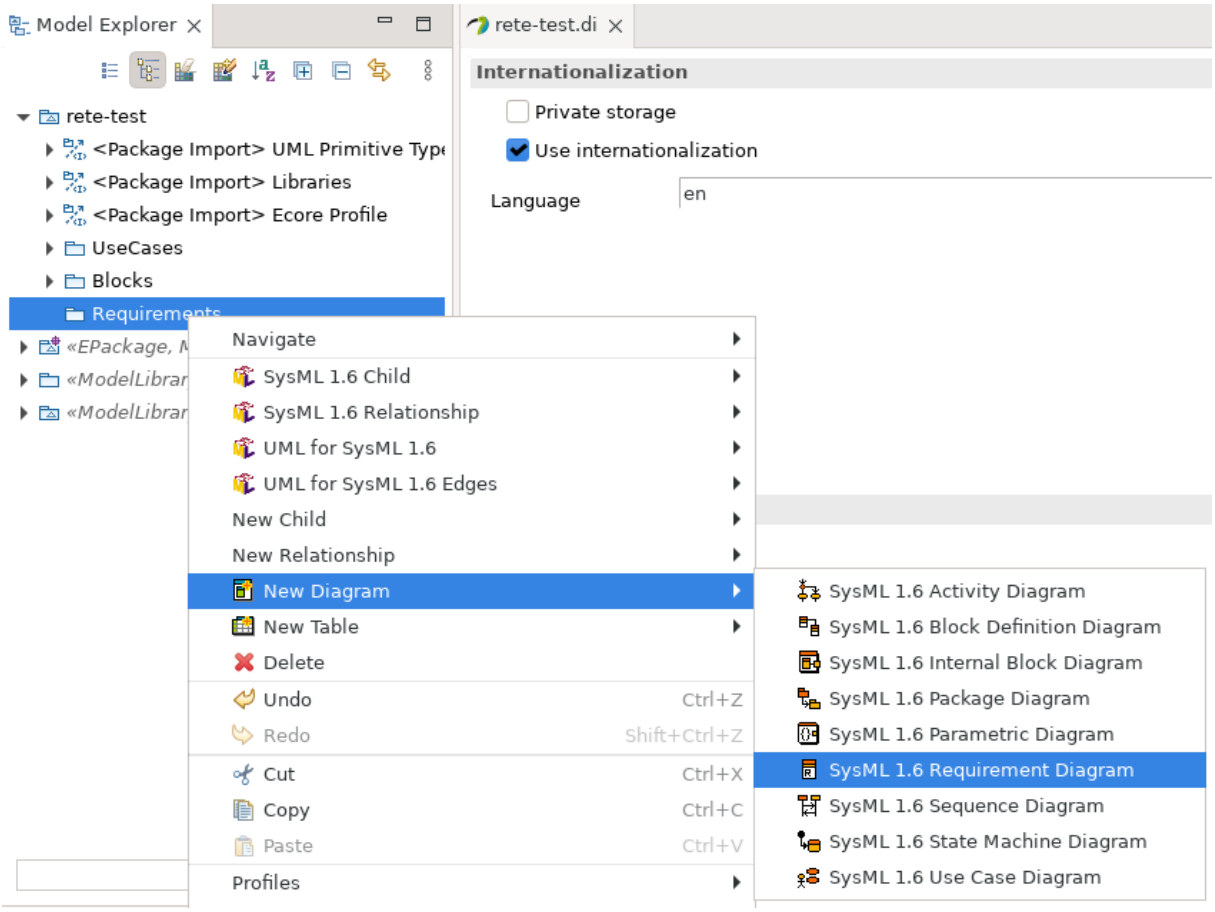
Requirement and Use case modeling Papyrus guidebook

Systems Engineering (BMEVIMIAC01)

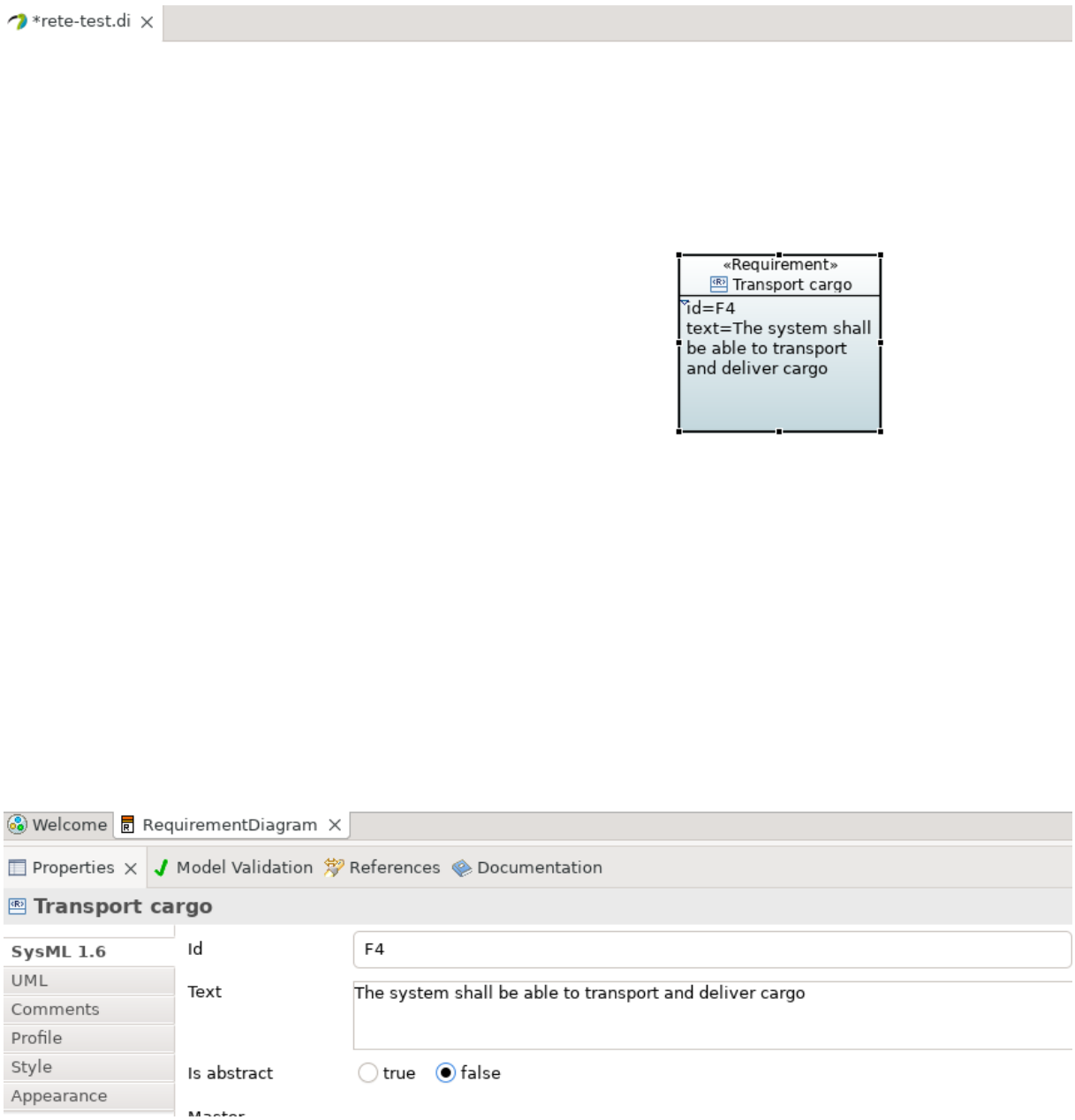
1 Requirement modeling

1.1 Diagram Creation

1. **Right click** on the selected *Package* (*Requirements* in this example) -> **New Diagram** -> **SysML 1.6 Requirement Diagram**.

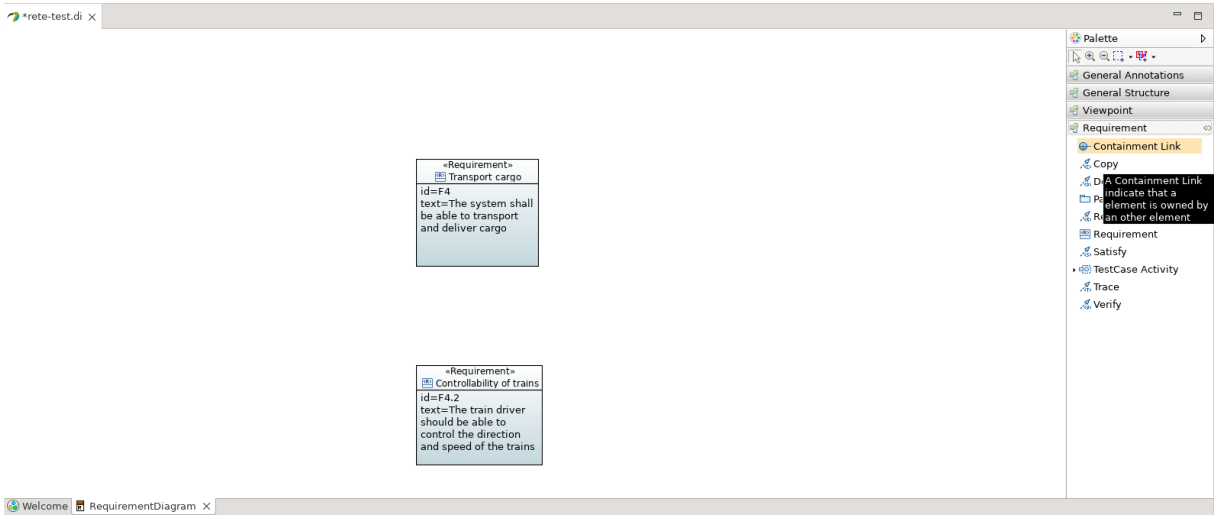


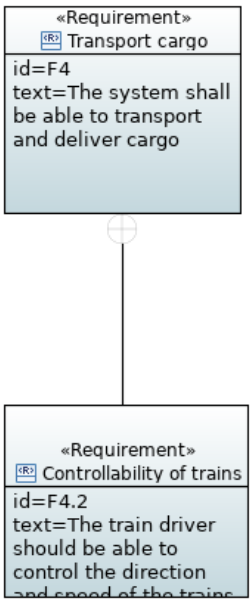
1. To add a *Requirement* select **Requirement** from the **Palette** and place it on the diagram, then on the **Properties** window, in the **SysML 1.6** tab fill out the **Id** and the **Text** fields.



1.2 Requirement Decomposition

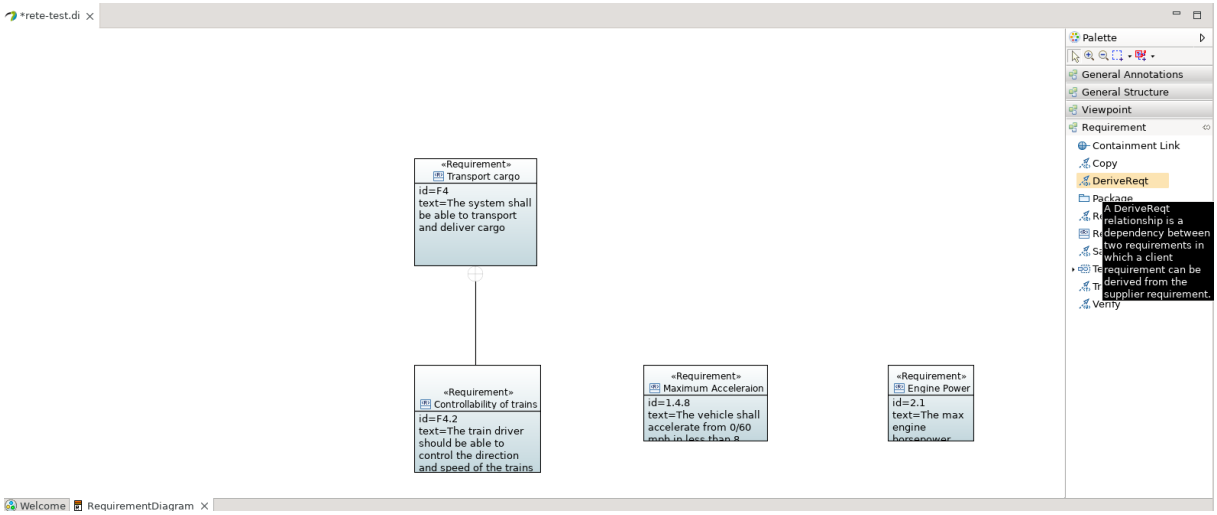
1. For *Requirement Decomposition* select **Containment Link** from the **Palette** and connect two *Requirements*. Pay attention to the direction of the connection!

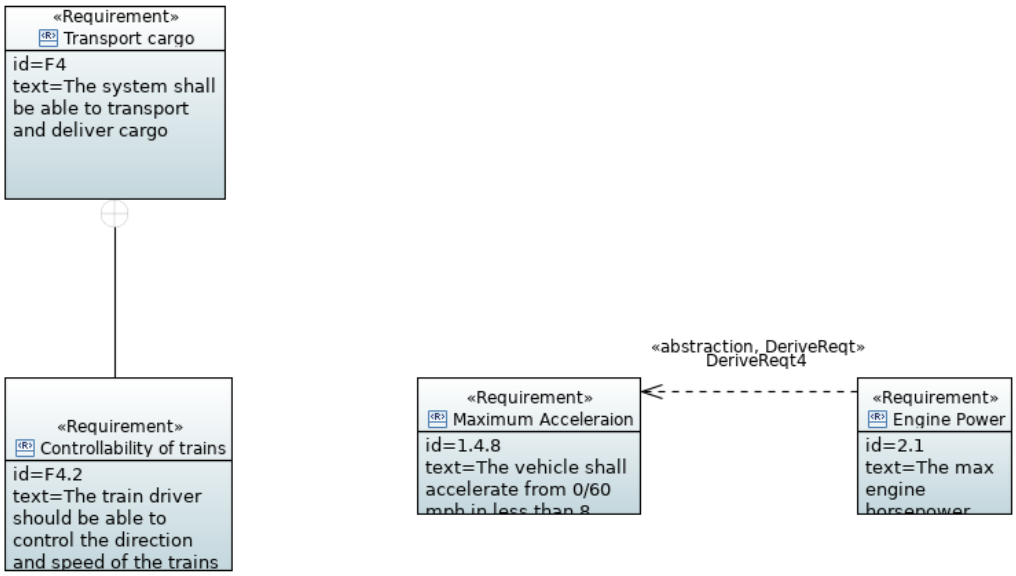




1.3 Deriving Requirements

1. To *Derive Requirements* select **DeriveReqt** from the **Palette** and connect two *Requirements*. Pay attention to the direction of the connection!

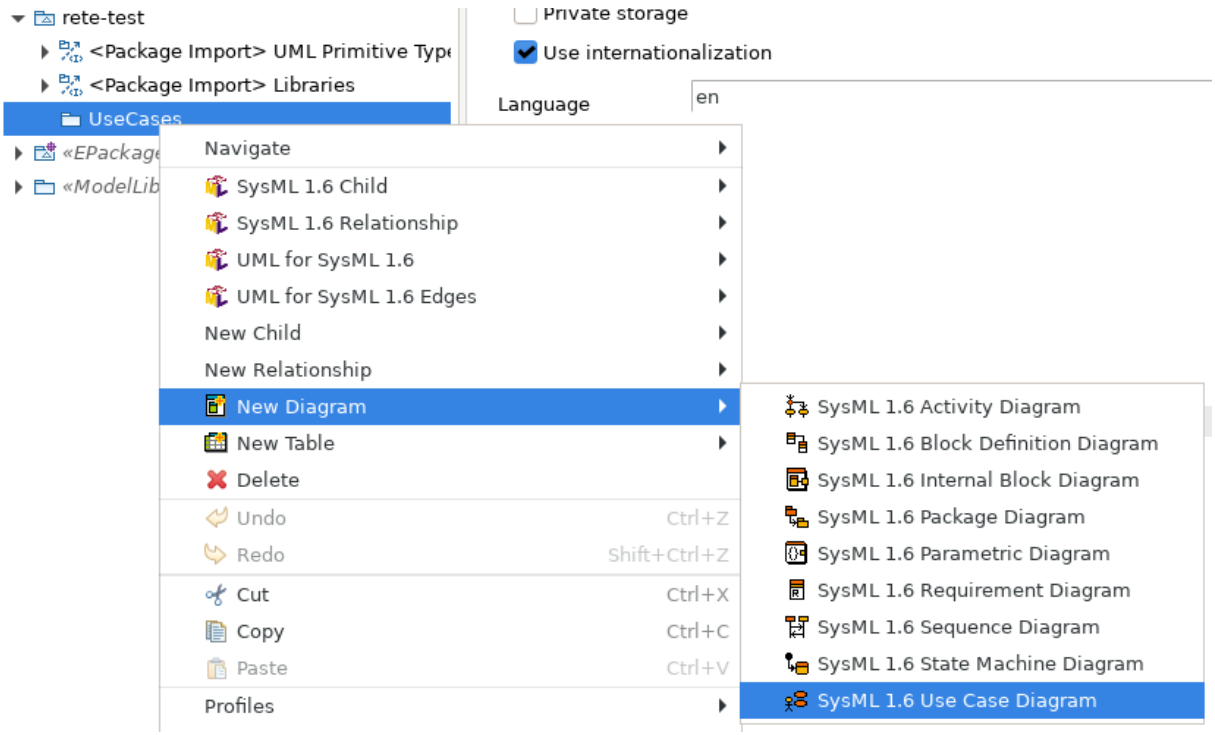




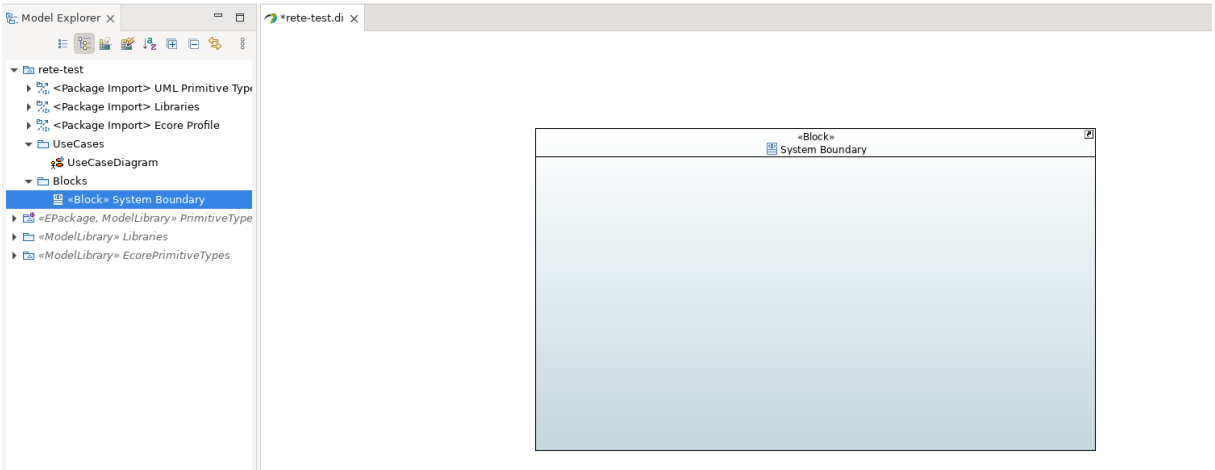
2 Use Case modeling

2.1 Diagram Creation

1. **Right click** on the selected *Package* (*UseCases* in this example) -> **New Diagram** -> **SysML 1.6 Use Case Diagram**.

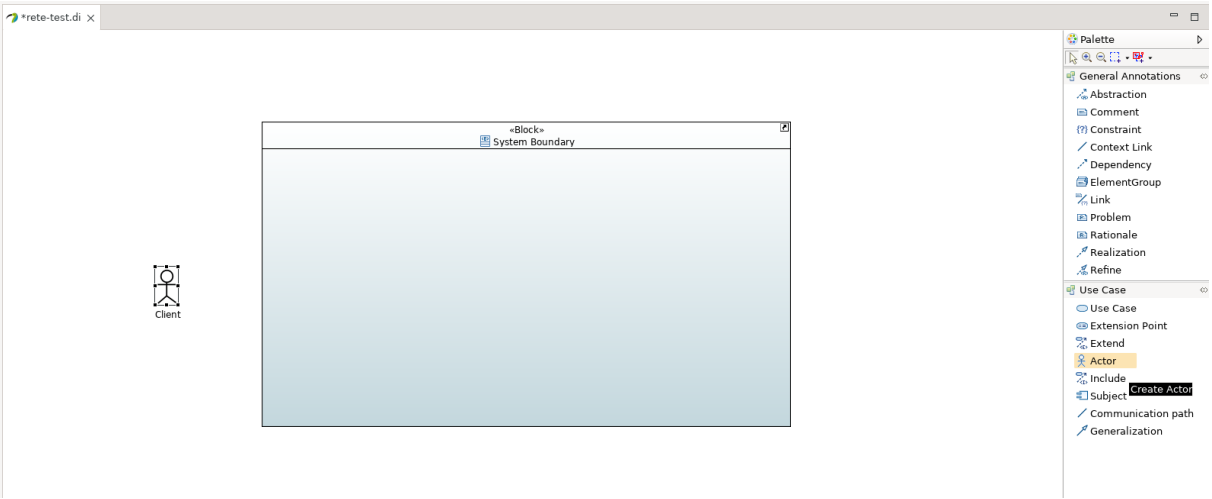


1. **Drag-and-drop** a block representing the boundry of the system (*System Boundary*) from the **Model Explorer** to the diagram.

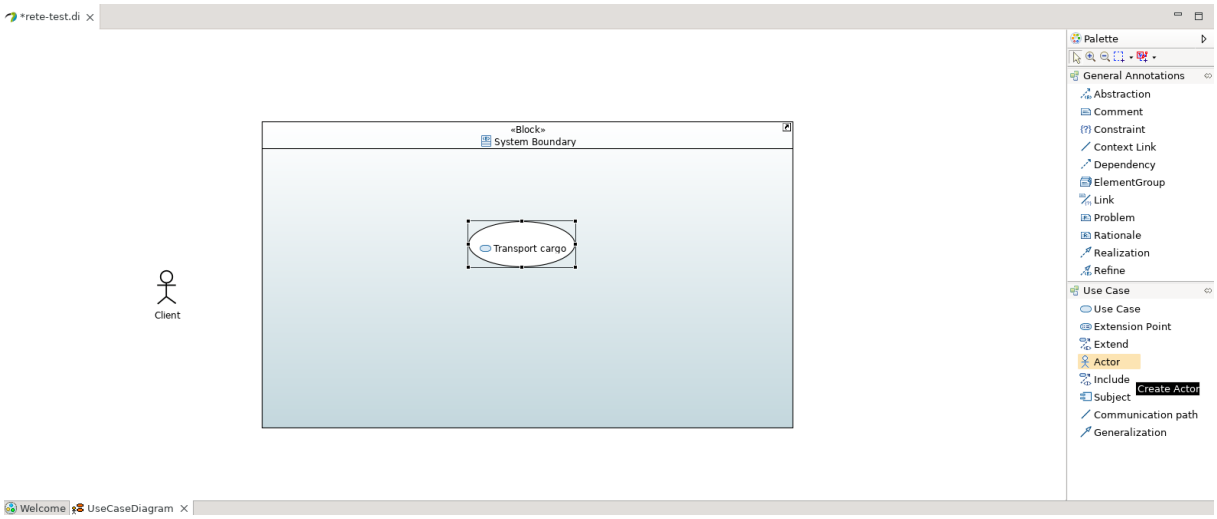


2.2 Adding Actors and Use Cases

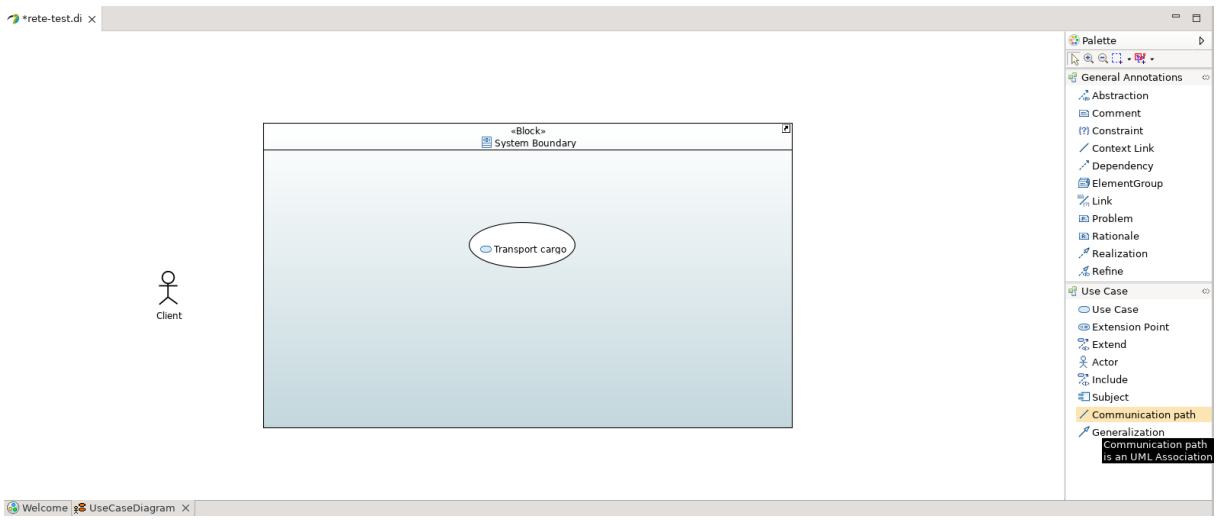
1. To add an *Actor* select **Actor** from the **Palette** and place it on the diagram.

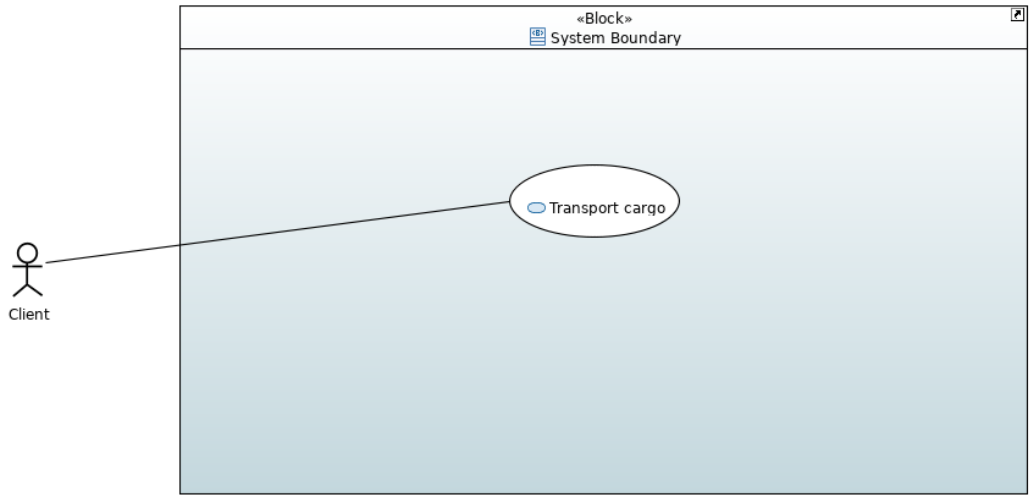


1. To add a *Use Case* select **Use Case** from the **Palette** and place it on the diagram.



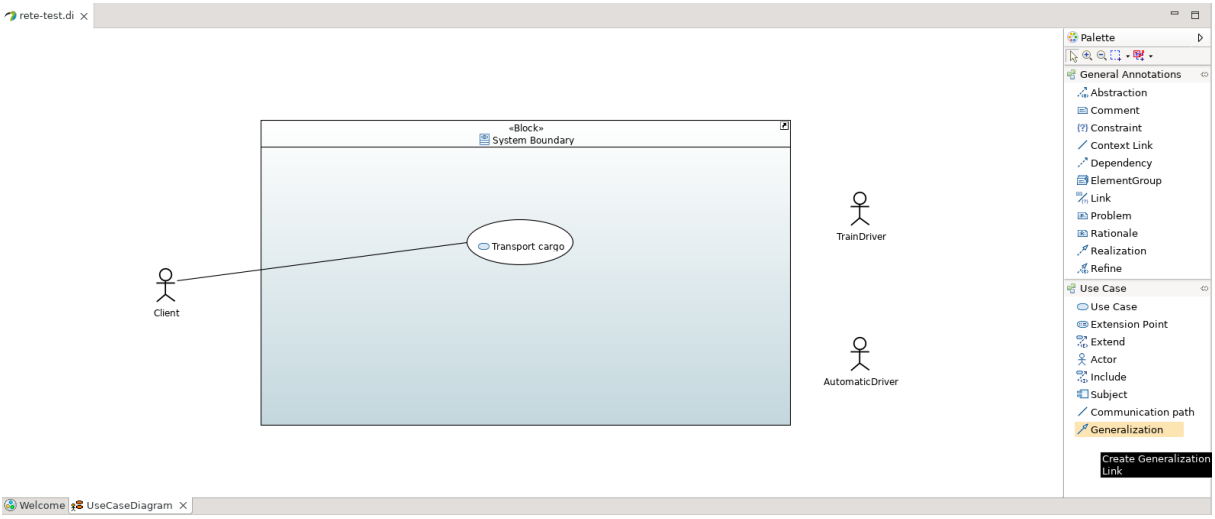
1. To add an *Association* between the *Actor* and the *Use Case* select **Communication Path** from the **Palette** and connect the *Actor* and the *Use Case*.

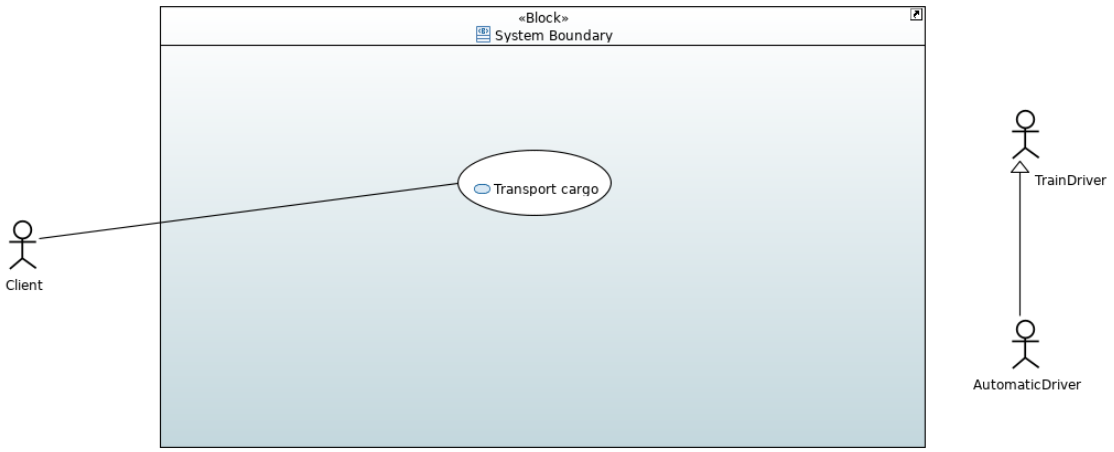




2.3 Relations between Actors

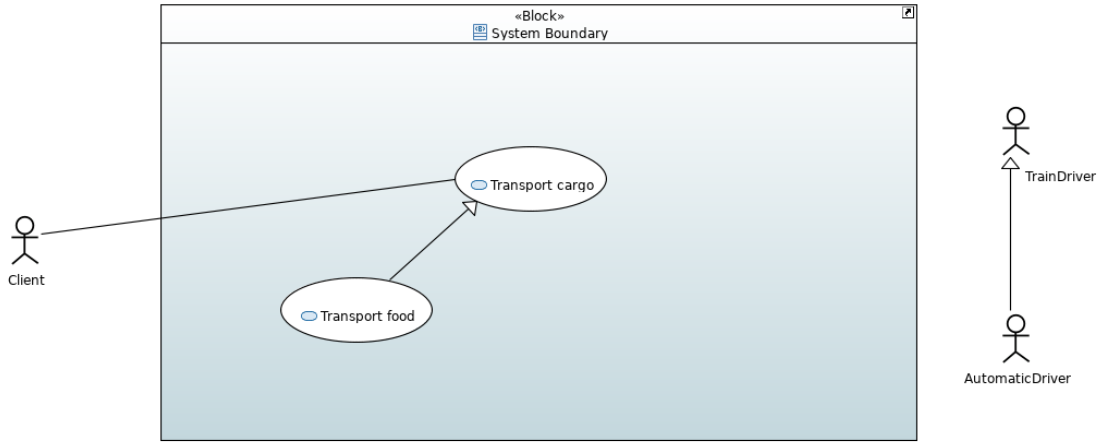
1. To *Generalize Actors* select **Generalization** from the **Palette** and connect two *Actors*. Pay attention to the direction of the connection!



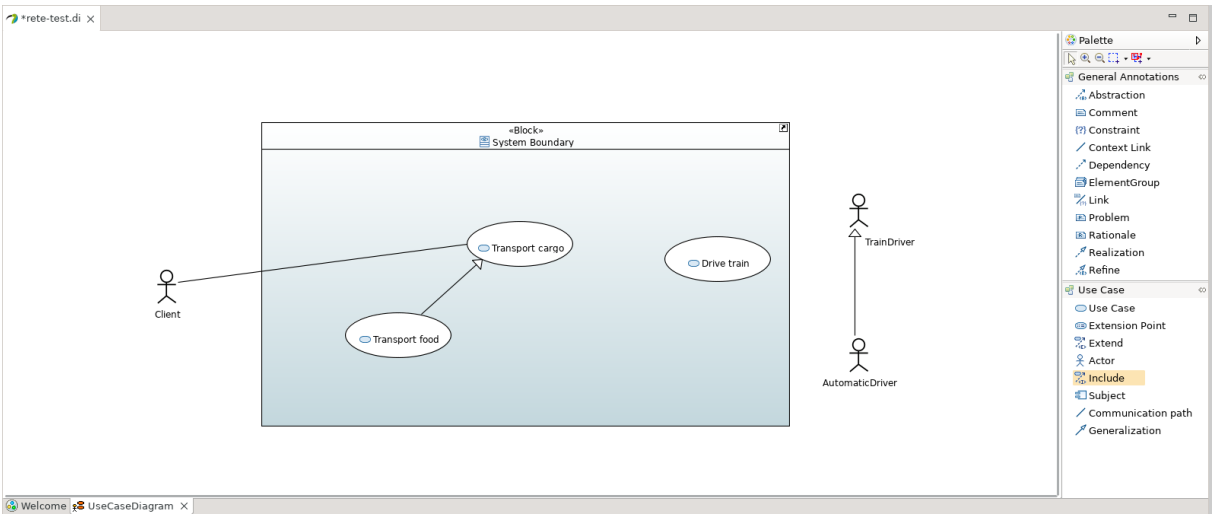


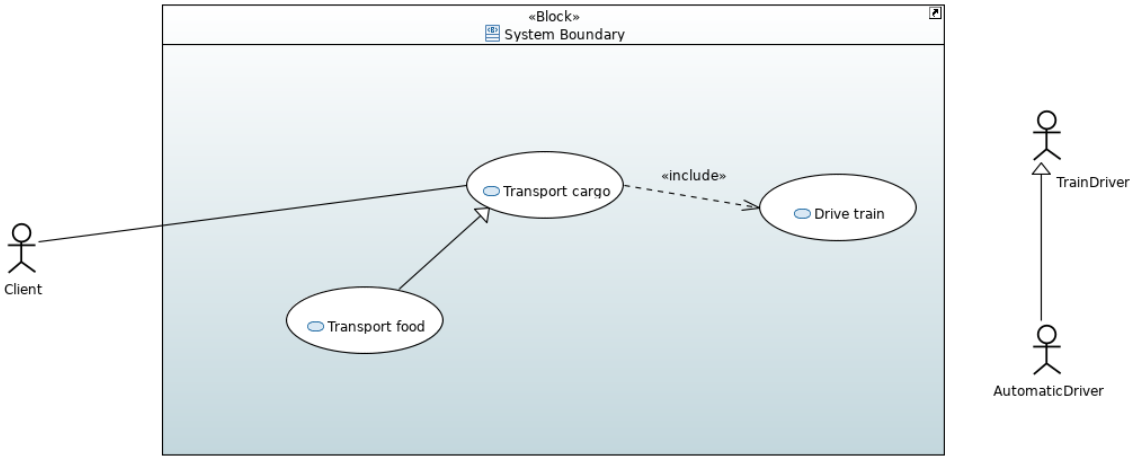
2.4 Relations between Use Cases

1. To *Generalize Use Cases* select **Generalization** from the **Palette** and connect two *Use Cases*. Pay attention to the direction of the connection!

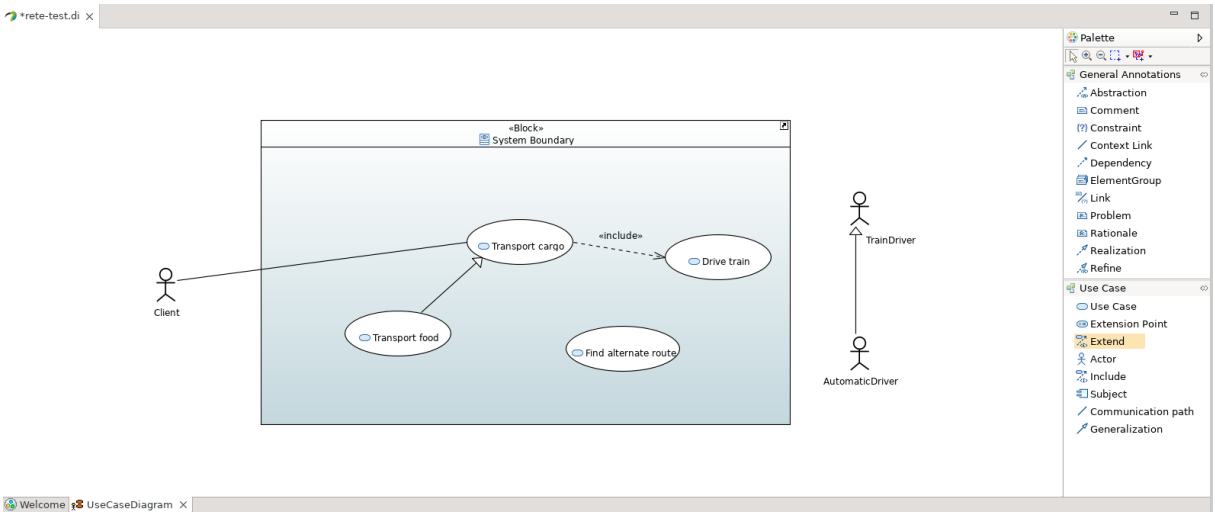


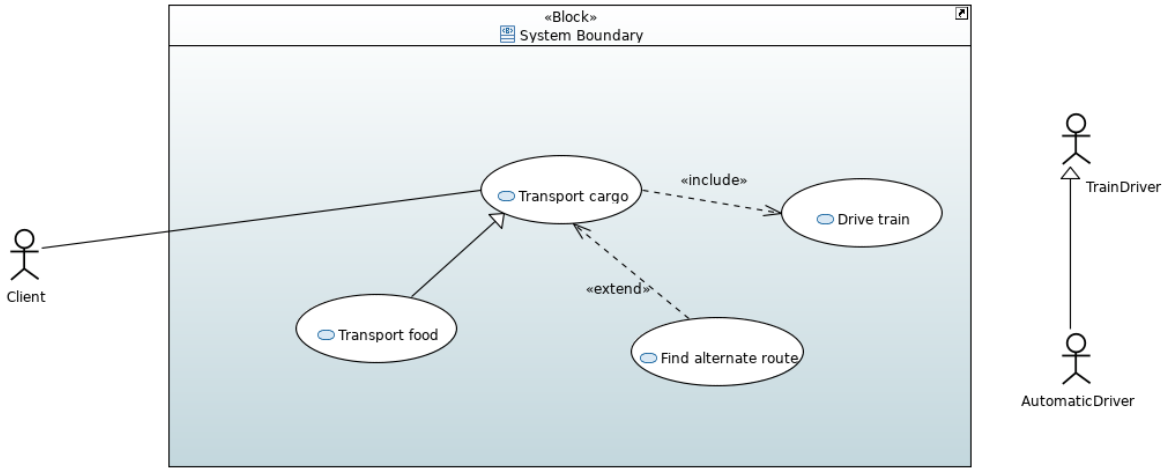
1. To *Refine* a *Use Case* select **Include** from the **Palette** and connect two *Use Cases*. Pay attention to the direction of the connection!





1. To *Extend* a *Use Case* select **Extend** from the **Palette** and connect two *Use Cases*. Pay attention to the direction of the connection!

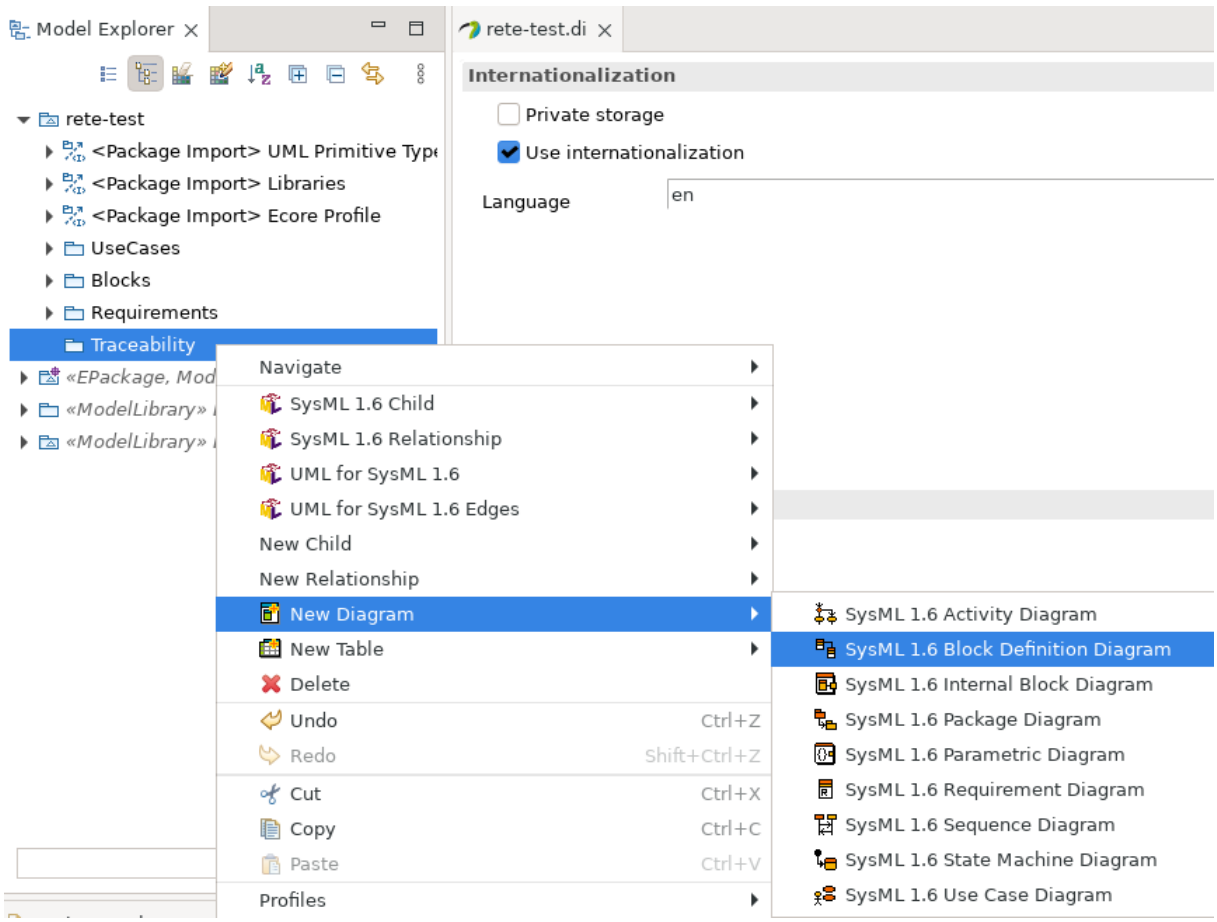




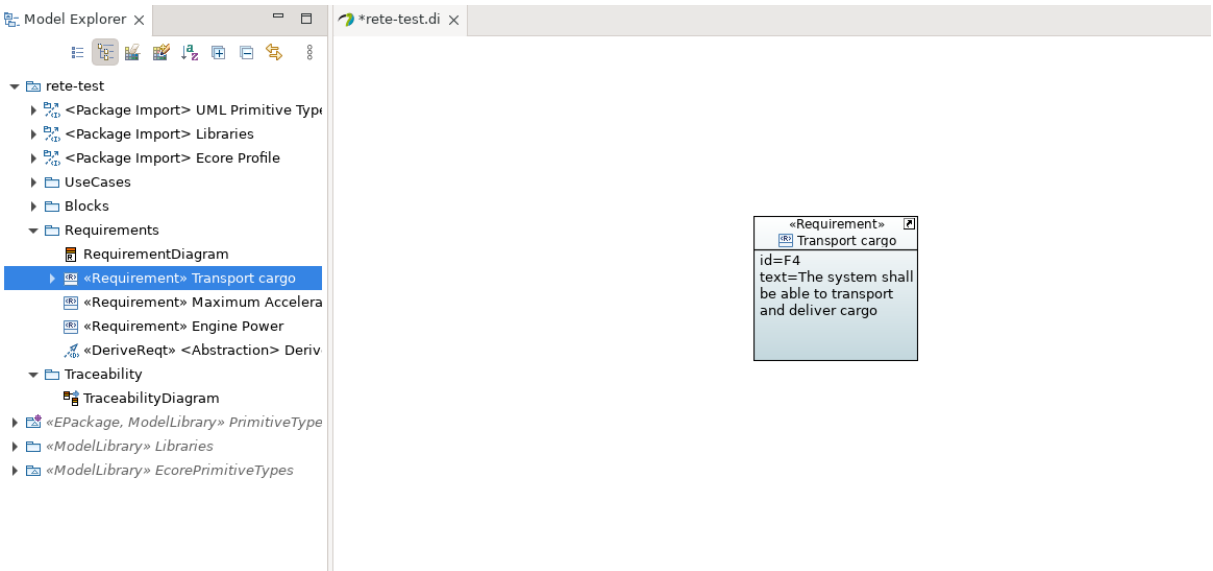
3 Traceability

3.1 Diagram Creation

1. **Right click** on the selected *Package* (*UseCases* in this example) -> **New Diagram** -> **SysML 1.6 Block Definition Diagram**.

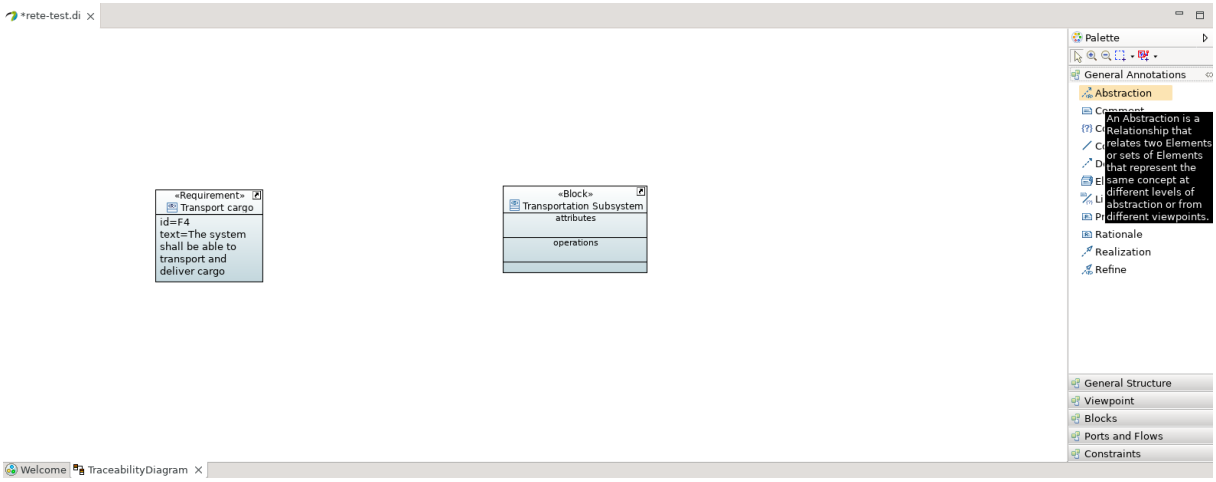


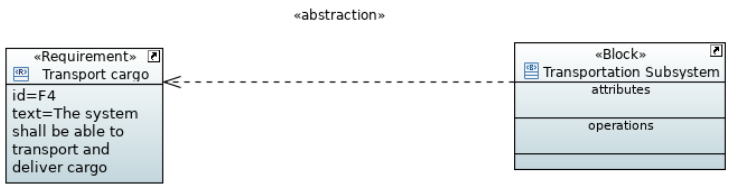
1. **Drag-and-drop** the *Requirement* (*Transport Cargo*) from the **Model Explorer** to the diagram.

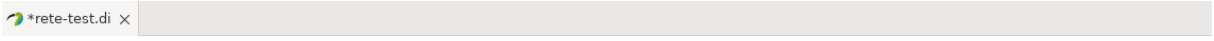


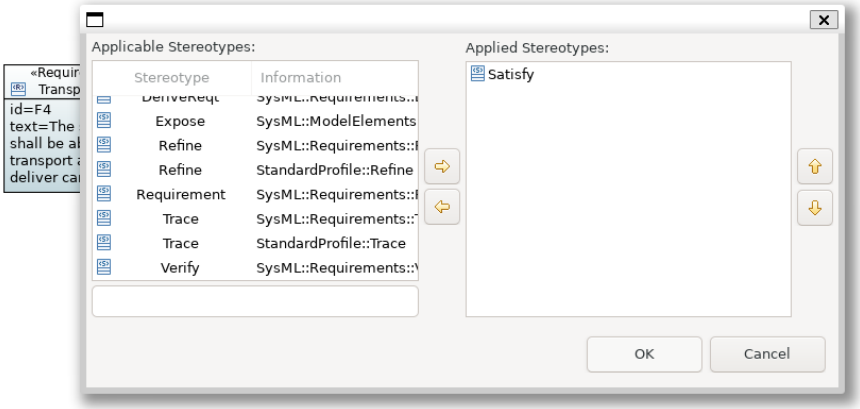
1. **Drag-and-drop** the *Block* satisfying the *Requirement* (*Transportation Subsystem*) from the **Model Explorer** to the diagram.

1. To *Satisfy* the *Requirement* select **Abstraction** from the **Palette** and connect the *Block* to the *Requirement*. Than on the **Properties** window, on the **Profile** tab add the **Satisfy** *Stereotype*-t. The **Satisfy** *Stereotype* must be in the **Applied Stereotypes** column!

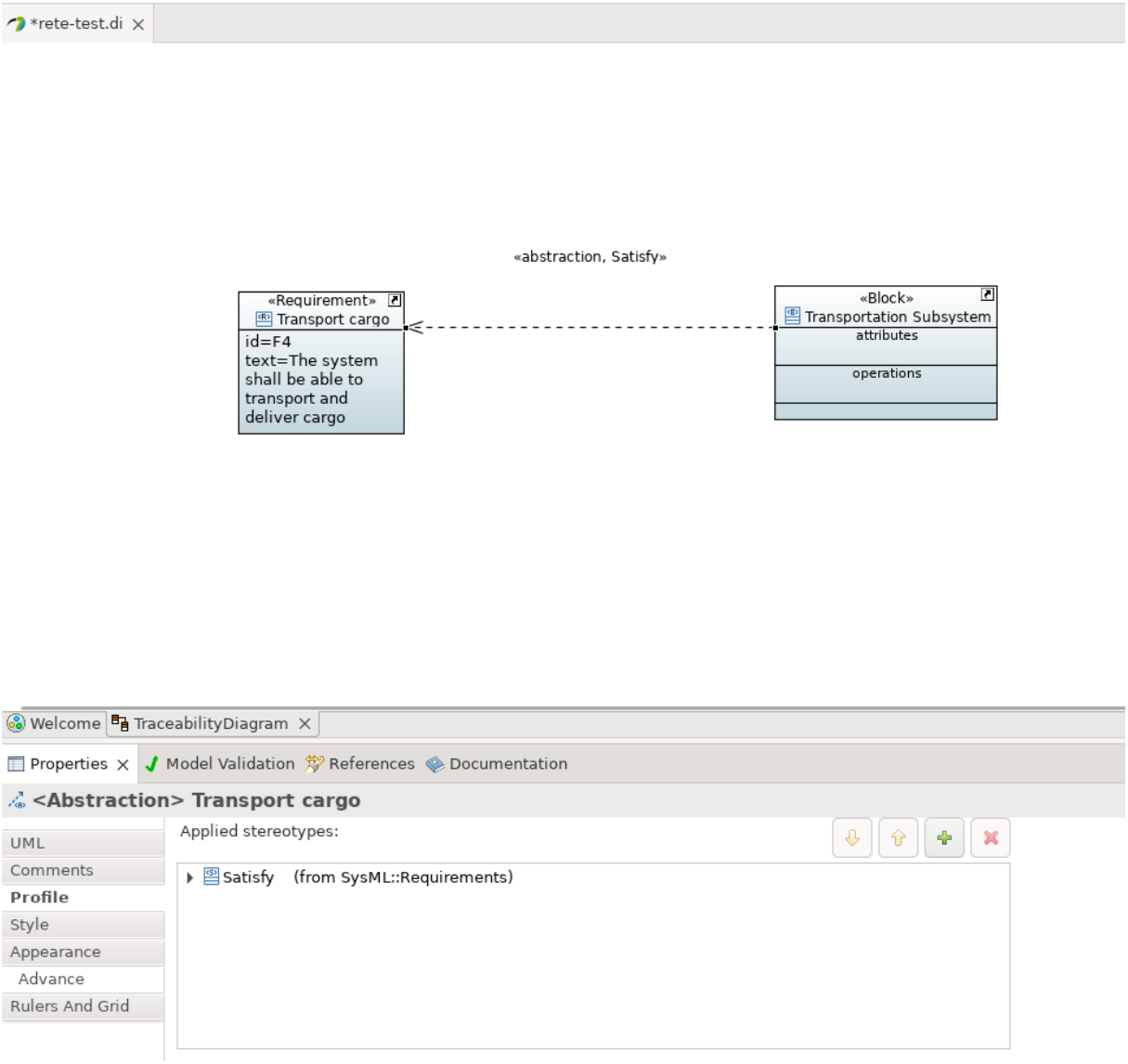



«abstraction»
«Requirement»
Transport cargo
id=F4
text=The system shall be able to transport and deliver cargo
«Block»
Transportation Subsystem
attributes
operations


*rete-test.di


Applicable Stereotypes:
Stereotype
Information
Expose
SysML::ModelElements
Refine
SysML::Requirements::
Refine
StandardProfile::Refine
Requirement
SysML::Requirements::
Trace
SysML::Requirements::
Trace
StandardProfile::Trace
Verify
SysML::Requirements::
Applied Stereotypes:
Satisfy
OK
Cancel

Welcome
TraceabilityDiagram
Properties
Model Validation
References
Documentation
<Abstraction> Transport cargo
UML
Comments
Profile
Style
Appearance
Advance
Rulers And Grid
Applied stereotypes:


*rete-test.di
«abstraction, Satisfy»
«Requirement»
Transport cargo
id=F4
text=The system shall be able to transport and deliver cargo
«Block»
Transportation Subsystem
attributes
operations
Welcome
TraceabilityDiagram
Properties
Model Validation
References
Documentation
<Abstraction> Transport cargo
UML
Comments
Profile
Style
Appearance
Advance
Rulers And Grid
Applied stereotypes:
Satisfy (from SysML::Requirements)